

PROJECT PROPOSAL

GROUP PROJECT

IIT 372-2

Group No: IIT 25

**Smart Commerce Core
(Learning Management System)**

Industrial Information Technology

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University of Sri Lanka

2026

DECLARATION

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Dr. J.A.V.M.K. Jayakody (vihara@uwu.ac.lk), Lecturer, Department of Computer Science and Informatics, Uva Wellassa University of Sri Lanka, and Mr. Y.E.Y Amarasekara (erandayapayasith@gmail.com), Demonstrator, Department of Computer Science and Informatics, Uva Wellassa University of Sri Lanka. This Project Proposal is submitted for the partial fulfilment of the requirement of the course unit IIT 372-2, Project II for the degree of Industrial Information Technology.

Group Details:

Group No: IIT 05

No.	Name of the Student	Index Number	E-mail address	Signature
1	K. Kavisana	UWU/IIT/22/090	iit22090@std.uwu.ac.lk	
2	A.H.M. Ayas	UWU/IIT/22/083	iit22083@std.uwu.ac.lk	
3	R.M.T. Madubhashini	UWU/IIT/22/081	iit22081@std.uwu.ac.lk	
4	A.G.P.P. Nandasiri	UWU/IIT/22/030	iit22030@std.uwu.ac.lk	

Supervisor Details:

Name of the Supervisor	E-mail	Contact Number
Dr. J.A.V.M.K. Jayakody	vihara@uwu.ac.lk	071 958 0948

Name of the Co-Supervisor	E-mail	Contact Number
Mr. Y.E.Y Amarasekara	erandayapayasith@gmail.com	076 528 9313

Project Approval:

.....

Dr. J.A.V.M.K. Jayakody

Date: 6/2/2026



.....

Mr. Y.E.Y Amarasekara

Date: 6/2/2026

Table of Contents

1. Introduction	1
1.1 Project Title	1
1.2 Project Description	1
1.3 Project Background and Motivation.....	2
1.4. Problem in Brief	3
1.5. Proposed Solution.....	3
1.6 Project aims and objectives	4
2. Methodology	5
2.1 Requirement Gathering and Analysis.....	5
2.2 System Design	5
2.3 Implementation.....	6
2.4 Testing and Debugging	6
3. Requirements Identification	9
3.1 Functional Requirements.....	9
3.2 Non-functional Requirements	11
3.3 System Requirements (Hardware / Software)	12
3.3.1. Software.....	12
3.3.2. Hardware Requirements	13
3.4 User Roles	13
3.5 User levels.....	16
4. Project Plan	17
5. References	18

Table of Figures

Figure 1: User Role Access Matrix Table..... 15

Figure 2: Gantt Chart 17

1. Introduction

1.1 Project Title

Smart Commerce Core (Learning Management System)

1.2 Project Description

Smart Commerce Core is a complete Learning Management System (LMS) made for institutes that focus on commerce education. It helps manage learning, administration, and finances all in one place, bringing together students, tutors, parents, and administrators.

The system allows:

- Teachers to upload lessons, quizzes, and assignments.
- Students sign up for courses, download materials, and submit their work.
- Parents to check student progress, attendance, view reports, and make payments.
- Admins to control the whole system, assign roles to teachers, and mark the attendance of students using QR code.
- It's built using JavaScript, PHP, HTML, CSS and uses a MySQL database to track grades, attendance, and payments. It can even predict and help solve academic problems early on.
- With built-in payment options and it shares news to parents and admins easier.

Overall, Smart Commerce Core is user-friendly, gives useful performance insights, and helps improve learning, making it a powerful tool for modern commerce education.

1.3 Project Background and Motivation

With the rapidly changing education world, especially in commerce, there is an increasing call for digital transformation. Conventionally, academic matters involving examinations, student lists, attendance, and feeing are cliché in their vocabulary. This is slow, error prone, and inefficient. Since many commerce academies still work with office-based/manual systems for admissions, marking, correspondence, etc., student engagement and academic performance at times can be negatively affected.

While the demand for individualized learning, real-time feedback, and seamless interaction among students, parents, tutors, and administrators is greater than ever, the degree of necessity of a centralized, smart solution has never been more. Yet, existing LMS usually cannot cater to the peculiarities of the workflows and account operations peculiar to commerce-oriented institutions and tuition centers.

The Smart Commerce Core - Smart Commerce Academy Portal fills this chasm. This project endeavors to design a role-based, integrated full-fledged academy management system pertinent to commerce education. For carrying out this much-needed automation of business processes, more efficient management of academic matters, and transparent communication among all stake holders dynamic content delivery using HTML, PHP, CSS and JavaScript, and strong data processing in MySQL.

Hence, while taking care of technical issues in education management, this solution is unable.

1.4. Problem in Brief

Most commerce academies still use old-fashioned systems that cause problems in teaching and managing the academy. Using manual methods can slow down feedback to students and create more work for tutors. This often leads to poor communication between teachers, students, and parents. As a result, students may miss deadlines, their performance may not be tracked properly, and they may lose interest in studying.

Parents also find it hard to stay updated on their child's progress or pay fees on time, because there is no central system with real-time information. Marking attendance using QR, helping students, and collecting fees also take a lot of time and can lead to mistakes.

All these issues affect student success, lower teacher motivation, and can harm the reputation of the academy. That's why there is a strong need for a central, digital system that helps manage academics, makes administration easier, and improves communication for everyone.

1.5. Proposed Solution

Smart Commerce Core envisions a fully web-centric Learning Management System specially tailored for commerce institutes. The system establishes the interface for academic, communication, and financial functionality by way of dashboards of functions for Admin, Tutors, Students, and Parents. Tutors can upload study materials, past lot, and model papers, make quizzes, and grade student performances through an auto-marking engine that is built with php logic. Students can handle course registrations, download resources, take quizzes, view their attendance. On the other hand, parents get to view the academic progress of their children, including attendance and grades, and pay fees securely via the integrated payment gateway.

Admins, however, are a higher level of supervisors for monitoring access, giving tutors access to subjects, and for generating reports in detail about student marks, attendance, and financial transactions. Running on a strong tech stack comprising HTML, CSS, JavaScript,

PHP, and MySQL, the system guarantees responsive front-end design, impeccable logic as per requirements, secure data handling for the institutions for decision-making.

1.6 Project aims and objectives

Project Aim

The aim of this project is to design and develop a centralized, user-friendly web-based platform tailored for a commerce tuition center, focusing on automating essential operations such as student registration, class scheduling, fee payment management, attendance tracking, academic performance monitoring, and facilitating effective communication between tutors, parents, and students.

Project Objectives

- To manage academic activities such as courses materials, quizzes, assignments, and attendance.
- To support real-time student performance tracking and progress reporting.
- To provide secure fee payment and enable parental access to student updates.
- To allow centralized admin control for user management, reporting, and notifications.

2. Methodology

Smart Commerce Core is being developed using the Waterfall Model of the Software Development Life Cycle (SDLC). This step-by-step method helps to make sure everything is done in the right order and of good quality. Each phase like planning, designing, building, testing, and launching is completed carefully before moving to the next.

By following this method, the team can meet current user needs and handle the technical environment well. This organized approach also sets up Smart Commerce Core for long-term success and user satisfaction.

2.1 Requirement Gathering and Analysis

The requirement gathering process involves identifying precise functional and nonfunctional requirements from administrators, tutors, students, and parents. Facts are currently being collected through:

- Interviews and questionnaires
- Analysis of the existing LMS system

The outcome of this phase is delivered in the form of a Software Requirements Specification (SRS) document, and this structured approach will continue to guide the system's development.

2.2 System Design

The system and user interfaces will be designed based on the collected requirements. The following will be the key design activities:

- MySQL Database Schema Designing: Structured relational data models will be created to efficiently store and manage system data.
- PHP Backend Design Planning: Designs will be prepared for server-side operations using PHP, including data processing, user authentication, and backend logic.

- Wire framing for Role-Based Dashboard and UI/UX: Wireframes will be developed for each user role Admin, Tutor, Student, and Parent to plan page layouts and navigation paths, ensuring a clear and smooth user experience.
- HTML will be considered during wire framing for structuring the page layouts and content hierarchy.
- CSS will be applied to ensure responsive styling, consistent theming, and visual appeal for all user roles, using stylesheets, Flexbox, Grid, and media queries.
- JavaScript will be used to make the UI interactive, enable real-time updates, and render user-specific content dynamically.
- ER Diagrams and Flowcharts will be created to illustrate entity relationships and data flow across various system modules.

2.3 Implementation

- The development phase will fetch system components using a combination of technologies and a modular design approach.
- PHP will handle all server-side processing, including communication with the database, session handling, and execution of business logic.

2.4 Testing and Debugging

Step-by-step testing will be performed to ensure the system is bug-free and robust:

- Unit testing will be conducted for individual modules.
- Integration Testing will be carried out to ensure proper interaction between modules.
- System Testing will be executed to validate the functionality of the complete platform.

- Bugs will be identified and resolved through an iterative process until stability is achieved.

2.5 Deployment

The system and user interfaces will be designed based on the requirements collected earlier.

The following will be the key design activities:

- **MySQL Database Schema Designing:** Structured relational data models will be created to efficiently store and manage system data.
- **PHP Backend Design Planning:** Designs will be prepared for server-side operations in PHP, including data processing, user authentication, and business logic.
- **Wire framing for Role-Based Dashboard and UI/UX:** Wireframes will be created for each role Admin, Tutor, Student, and Parent to plan page layouts and navigation paths. These wireframes will help ensure clear and smooth user experience.
- **HTML** will be used during wire framing to define page structure and layout.
- **CSS** will be applied for responsive styling, visual consistency, and appealing theming across all roles, using stylesheets, Flexbox, Grid, and media queries.
- **JavaScript** will be utilized to make the UI interactive, enable real-time updates, and render user-specific content.
- **ER Diagrams and Flowcharts** will be developed to illustrate entity relationships and data flow across system modules.

2.6 Maintenance

After deployment, the system will be maintained to ensure stability, security, and smooth performance. The following activities will be carried out:

- Fixing bugs and technical issues
- Updating and improving system features
- Performing database backup and maintenance
- Monitoring system performance
- Providing user support.

This phase helps Smart Commerce Core remain secure, reliable, and efficient for long term use.

3. Requirements Identification

3.1 Functional Requirements

User Login

- Users (Admins, Tutors, Students, Parents) should be able to register and login.
- The system should show different dashboards for each user type (Admin, Tutor, Student, Parent).

Course and Material Management

- Admins can create, update, and manage courses within the system.
- Tutors can upload and organize study materials such as notes, PDFs and past papers.
- Students can join courses and view or download learning materials anytime.
- The system can recommend study materials based on student performance and learning progress.
- The system can suggest the next topic for students to learn according to their completed lessons and quiz results.

Assignments and Quizzes

- Tutors can create quizzes with auto-grading functionality.
- Students can attempt quizzes online and view their marks instantly after submission.
- Tutors can upload assignments, review student submissions, and provide feedback and grades.

Attendance Checking

- Admin can mark student attendance using QR code scanning.
- Students and parents can view attendance records through the system.
- The system can display attendance trends to identify whether student attendance is improving or decreasing over time.

- Parents can receive low attendance alerts and notifications when a student's attendance falls below the required level.

Performance Reports

- The system should display student performance using graphs and analytics to identify strong and weak subject areas.
- Performance reports should be accessible to students, tutors, and parents.
- The system can analyze performance trends to identify whether student results are improving or declining over time.
- The system can detect weak topics or subjects where students require additional support.
- The system can provide at-risk student indicators to help tutors and parents identify students who may need academic assistance.

Notifications

- Students and parents should receive notifications for quizzes, assignments, class schedules, and payments.
- Tutors should receive reminders to upload grades and study materials on time.
- The system can provide smart reminders based on student activities, such as low quiz scores, pending assignments, or upcoming deadlines.

Payments and Finance

- Parents can upload payment receipts through the system.
- Admin can verify and accept payments after checking uploaded receipts.

3.2 Non-functional Requirements

Performance

- The system should load pages and respond to actions quickly (within 2 seconds).
- Quiz results should be shown right after submission.

Security

- Passwords should be safely stored using password hashing (like bcrypt).
- The system should use HTTPS to keep data safe online.
- Only the right users should be able to access specific data (like students can't access admin pages).

Usability

- The system should be easy to use for everyone. The system should provide an intuitive and user-friendly interface accessible to users of all roles and technical backgrounds.
- It should work well on both computers and mobile phones.

Availability

- The system should work 24/7 with very little downtime.
- Important data should be automatically saved (backed up) every 12 hours.

Scalability

- The system should support many users at the same time.
- It should be easy to add new users, courses.

Maintainability

- The code should be written in a clean way so it's easy to fix bugs and update the system.

3.3 System Requirements (Hardware / Software)

3.3.1. Software

- **Development Environment:** Visual Studio Code will be utilized as the principal IDE for frontend (html) and backend (PHP).
- **Database Management System:** MySQL will be the relational database system to handle student, user, and financial data.
- **Web Server:** XAMPP will be employed to run a local server environment in order to test PHP-MySQL web applications.
- **Version Control:** Version control and collaboration will be done using Git and GitHub.
- **Project Management:** Trello or Asana will be used for task assignment, sprint planning, and monitoring.
- **Documentation:** Microsoft Word / Microsoft PowerPoint
MS Word is used to prepare formal documentation including project reports.
MS PowerPoint is used to create project presentations and visual explanations.
specifications.

Backend: PHP

- Vanilla PHP will be employed for flexibility and direct access to the codebase, enabling tailored features for educational needs.
- Development time will be longer than in frameworks such as Laravel or Symfony but will provide better optimization for logic-specific features such as grading schemas.
- PHP will be natively integrated with MySQL for effective data operations.

Frontend: JavaScript, HTML, CSS

- The frontend will be built using HTML to achieve responsiveness and role specific UI experiences for students, tutors, parents, and admins.

- Its component-based architecture and virtual DOM will allow scalable and fast updates to the UI.

Database: MySQL

- MySQL will be used as a robust relational database system that is suitable for structured academic data such as grades, attendance, and finance.
- In-built PHP support for MySQL will provide secure and effective storage and retrieval of data.

3.3.2. Hardware Requirements

Laptop / Desktop Computer

- Minimum 4GB RAM
- Minimum 10GB of free disk space
- Dual-core or higher processor (Intel i3 or above recommended)
- Operating System: Windows 10 or newer

Network Connectivity

- Stable Internet connection with Wi-Fi adapter or Ethernet port

3.4 User Roles

Admin

- Approves and manages tutor, student, and parent accounts.
- Creates and manages courses, assigns tutors, and sets up academic calendars.
- Monitors attendance using QR code, grades, and payments.
- Generate reports for academic performance and financial tracking.
- Deactivates student accounts with overdue payments.

Tutor

- Uploads study materials like notes, past papers.
- Creates and manages quizzes and assignments with auto-grading features.
- Views student performance and provides feedback.

- Communicates with students and receives reminders to update grades/materials.

Student

- Registers for courses, downloads materials, and submits assignments.
- Takes quizzes and gets instant results.
- Views grades, attendance, and performance reports.
- Receives notifications for upcoming tasks and fee status.

Parent

- Monitors child's grades, attendance, and academic progress.
- Tracks assignment submissions and quiz attempts.
- Receives fee payment reminders and make payments online.
- Gets real-time notifications about academic performance and deadlines.

User Role Access Matrix

No table of figures entries found.	Student	Tutor	Parent	Admin
View Courses	✓	✓	✓	✓
Enroll in Courses	✓			✓
Submit Assignment	✓			
Download Study Materials	✓	✓		✓
Upload Study Materials		✓		✓
Design Quizzes		✓		
Attempt Quizzes	✓			
View Grades	✓	✓	✓	✓
View Attendance	✓	✓	✓	✓
Automated Grading		✓		✓
Predictive Performance Reports		✓	✓	✓
Fee Payment	✓		✓	✓
Manage User Accounts				✓
Register/Login	✓	✓	✓	✓
Configure System Settings				✓
Recommend Study Materials	✓			✓
Suggesting Next Topic	✓			✓

Figure 1: User Role Access Matrix Table

✓ Indicates the user role has permission for the corresponding feature.

3.5 User levels

Admin

- Manage student, tutor, and parent accounts.
- Assign subjects and classes to tutors.
- Create and manage courses.
- Generate reports (attendance, grades, financials).
- Manage the fee system and track payments.
- Configure and update system settings.
- Post announcements and system notices.
- Mark student attendance using QR code.

Tutor

- Upload study materials (PDFs, notes).
- Create and manage quizzes using PHP-based auto-grading.
- Mark assignments and give feedback.
- Communicate with students through messages/forums.
- Monitor student performance.

Student

- Attend classes (physically or virtually).
- Download and view study materials.
- Attempt quizzes and submit assignments (Assignments are tasks given by tutors that students complete and submit online for evaluation).
- View attendance, grades, and performance reports.
- Receive notifications and alerts.
- Making fee payments

Parent

- Track child's academic performance and attendance.
- View grades, assignment submissions, and quiz attempts.
- Upload the payment slip of the student.
- Receive important notifications/reminders.

4. Project Plan

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Identify topic, gathering requirements															
Requirements Analysis & specification															
Database design															
Interface design															
Development															
Testing and debugging															
Verification															
System implementation															
System Testing															
Documentation															

Figure 2: Gantt Chart

5. References

- [1] GitHub. Introduction to Git and Version Control. GitHub Docs. Retrieved June 2025 from <https://docs.github.com>

- [2] W3Schools. HTML, CSS, MySQL Tutorials. Retrieved June 2025 from <https://www.w3schools.com>

- [3] Oracle Corporation. MySQL 8.0 Reference Manual. Retrieved June 2025 from <https://dev.mysql.com/doc/>

- [4] Code Astro. Complete Online Learning Management System in PHP MySQL. Retrieved June 2025 from <https://codeastro.com/complete-online-learning-management-system-in-php-mysql/>